2.3.3.1 Step-up Transformer

In this type, the number of turns in secondary coil must be more than the turns in the primary coil. These types of transformers are used in power generating plants, where the electricity is carried over a long distance and also in Television. The step up transformer is shown in Figure 2.16 and its turns ratio is given in Table 2.1.

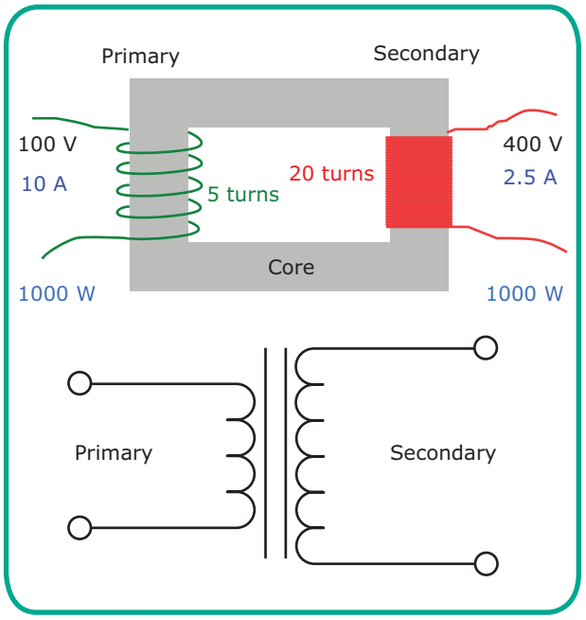


Figure 2.16 Step-up Transformer

Table 2.1 Turns ratio of step-up transformer

	Primary	Secondary
Turns ratio	*1*	*2*
Voltage ratio	*1*	*2*

2.3.3.2 Step-down Transformer

In this type, the number of turns in the secondary coil must be less than the number of turns in the primary coil. This type of transformer is used in Radio and TV receivers, eliminators and other video equipment. Figure 2.17 shows a step-down transformer. Table 2.2 summarizes the turns ratio of step-down transformer.

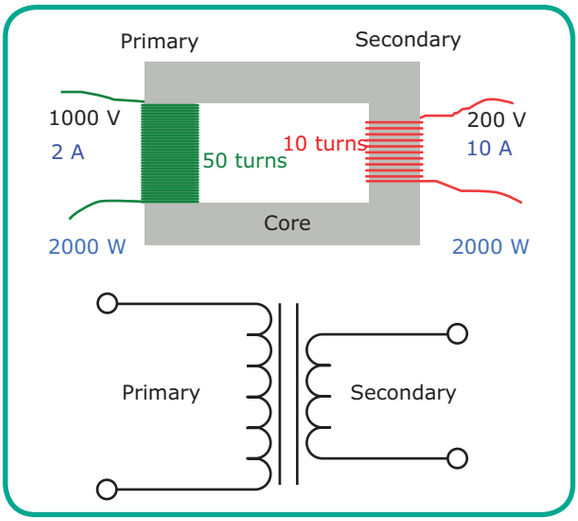


Figure 2.17 Step-down Transformer

Table 2.2 Turns ratio of step-down transformer

	Primary	Secondary
Turns ratio	*2*	*1*
Voltage ratio	*2*	*1*

2.3.3.3 Isolating Transformer

This type of transformer has 1:1 ratio of winding. These types of transformer are used mainly in fault finding places. This transformer is used to avoid materials to have direct contact with main supply. Figure 2.18 shows an isolation transformer.

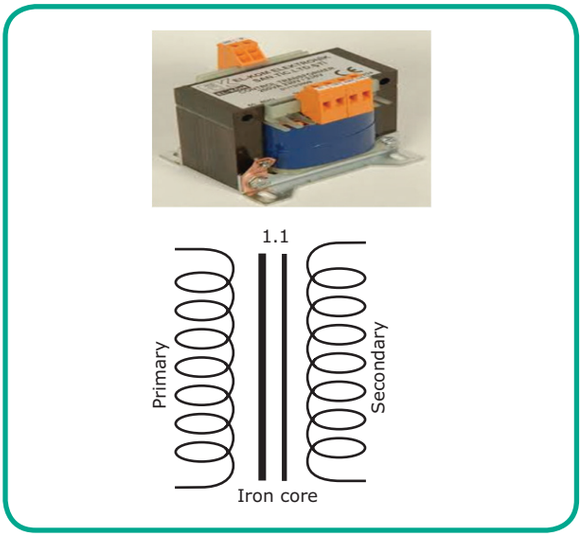


Figure 2.18 Isolating Transformer

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